

Data Science Hackathon 2025

Group 5

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Introduction

In astronomical imaging, detecting and analyzing sources accurately is a big challenge, especially in radio frequencies. Raw images contain significant noise and artifacts, which can obscure important astronomical sources and affect scientific analysis.

PROBLEM

One of the key issues is how to **effectively clean and preprocess data** without losing valuable information. Techniques like **sigma clipping** help remove noise, but if applied incorrectly, they can also remove real astronomical sources

Objective

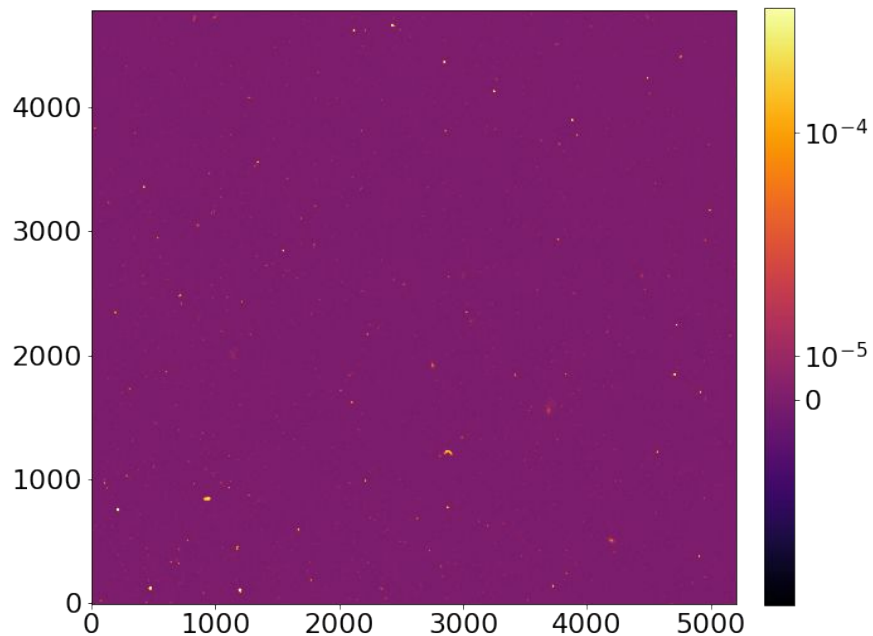


1. Analyze FITS images at different frequencies; 1400 MHz, 560 MHz, 9200 MHz
2. Improve the training pipeline
3. Compare ML models across frequencies

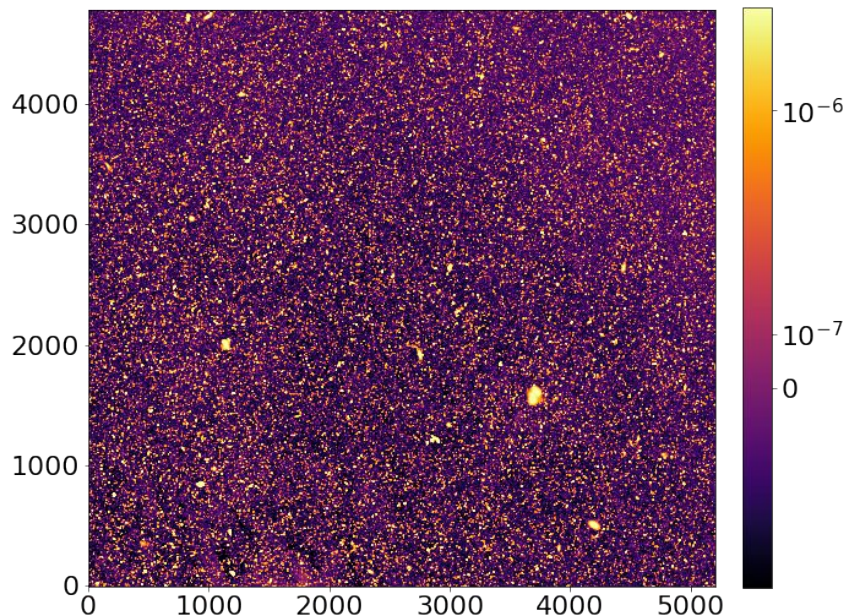


Preprocessing Overview & Results

-Before Sigma Clipping (560MHz)



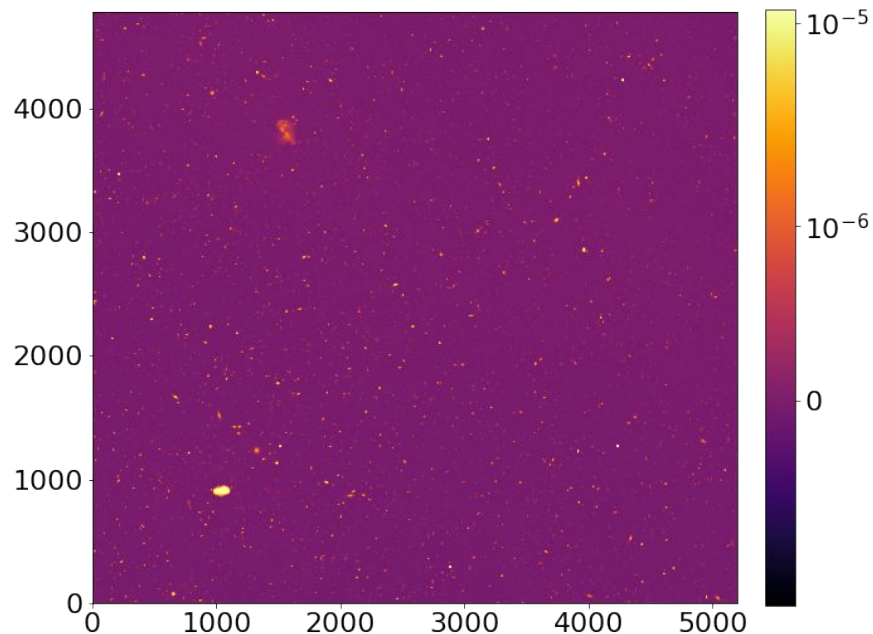
-After Sigma Clipping (560MHz)



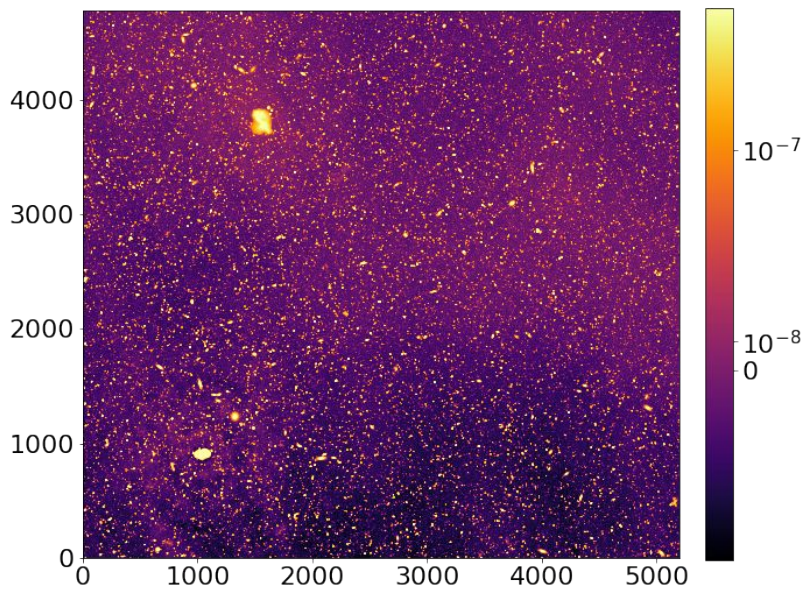


Preprocessing Overview & Results

-Before Sigma Clipping (1400MHz)



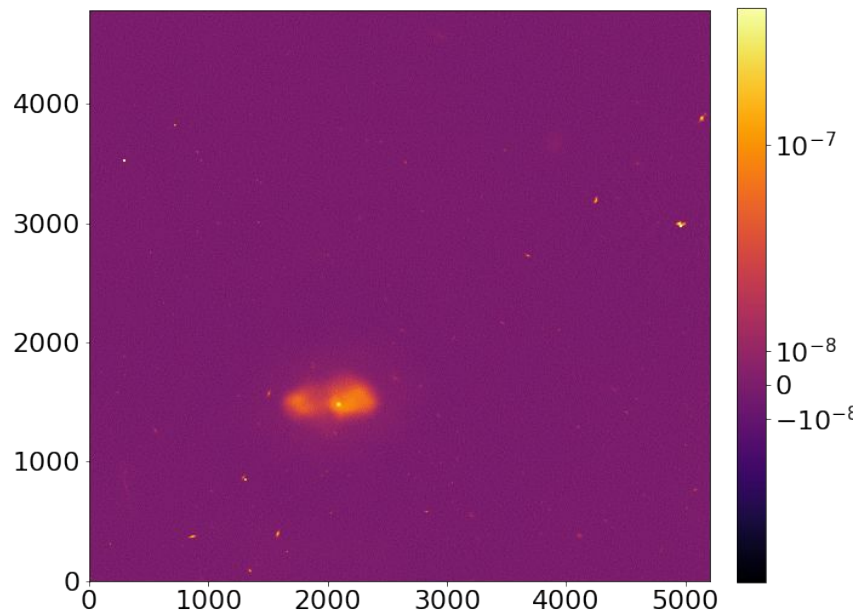
-After Sigma Clipping (1400MHz)



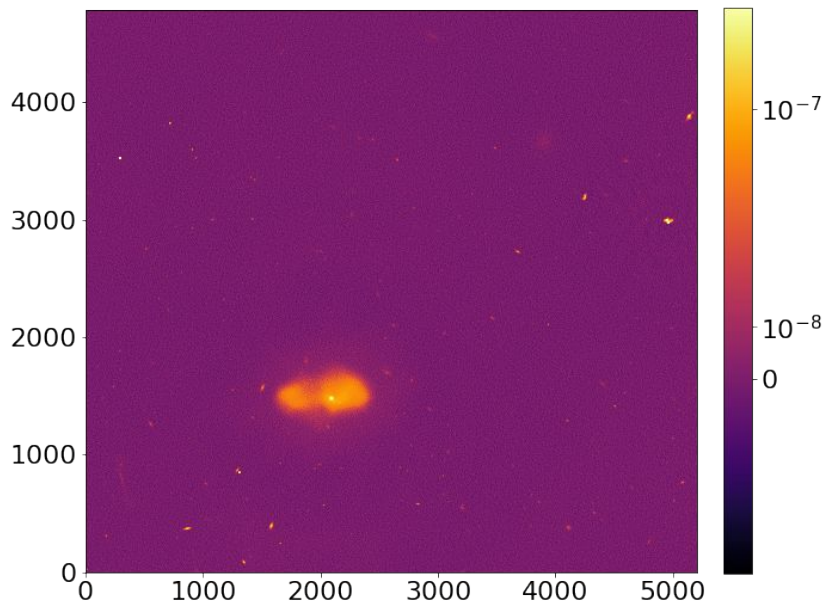


Preprocessing Overview & Results

-Before Sigma Clipping (9200MHz)

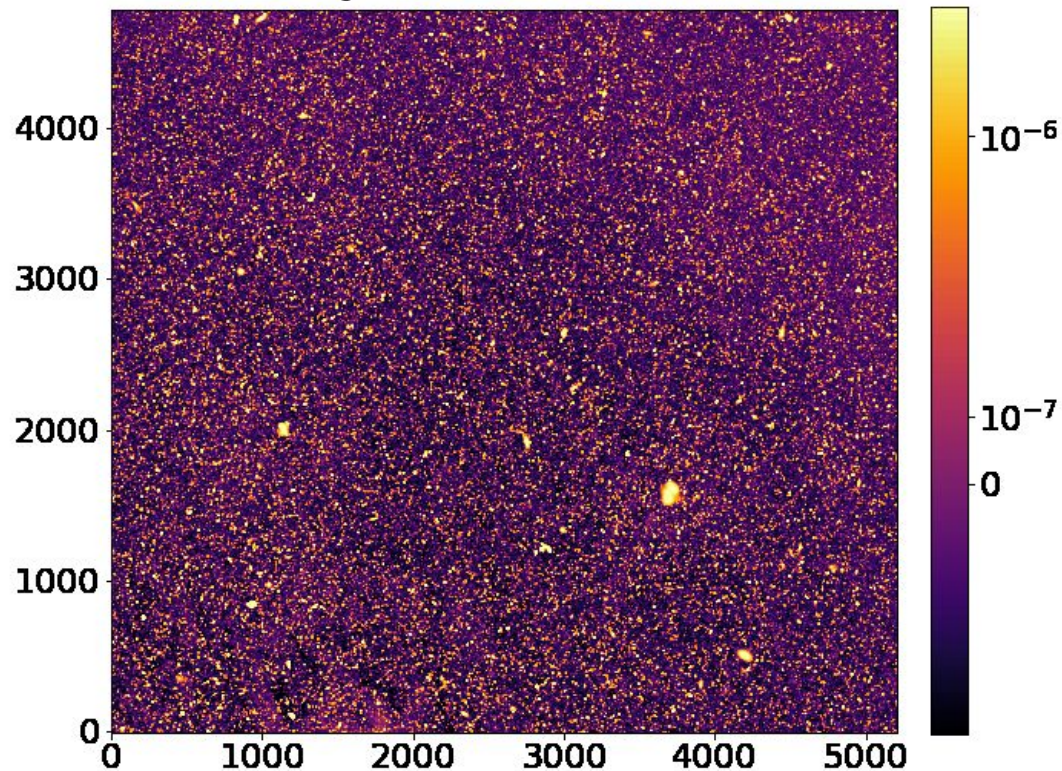


-After Sigma Clipping (9200MHz)



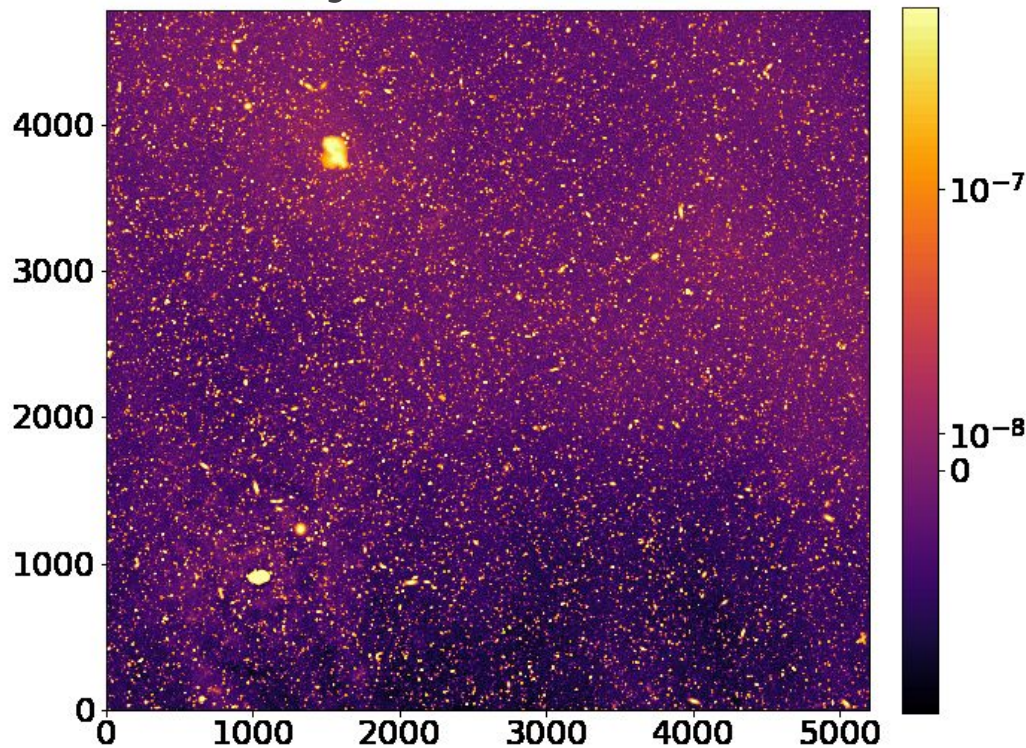
Preprocessing Overview & Results

-Primary Beam Correction GIF(560MHz)



Preprocessing Overview & Results

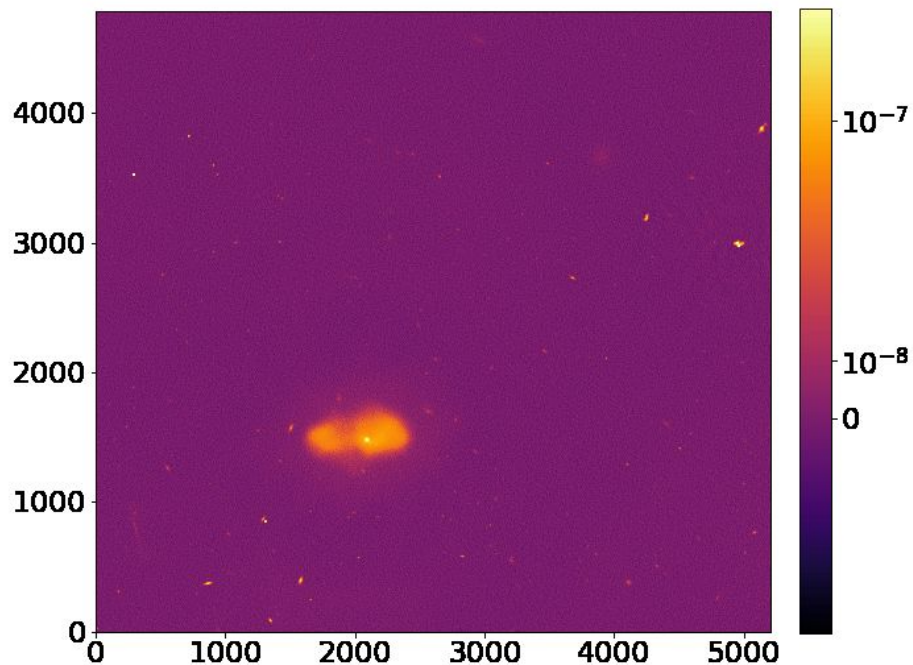
-Primary Beam Correction GIF(1400MHz)





Preprocessing Overview & Results

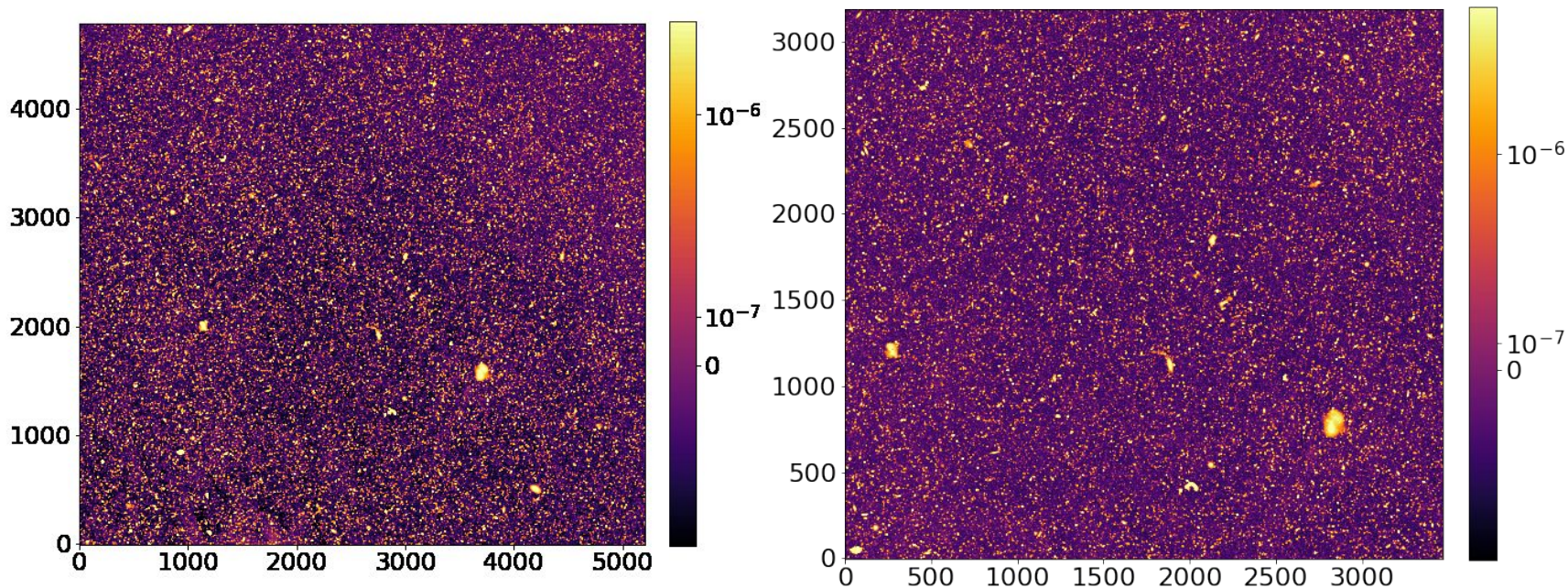
-Primary Beam Correction GIF(9200MHz)





Preprocessing Overview & Results

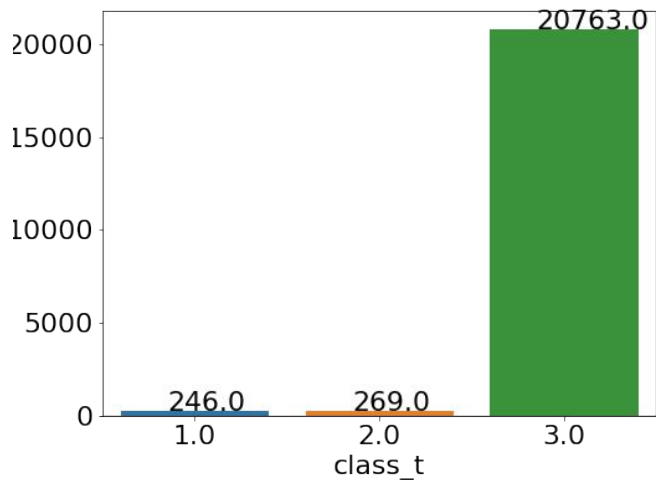
-Training Image(560MHz)



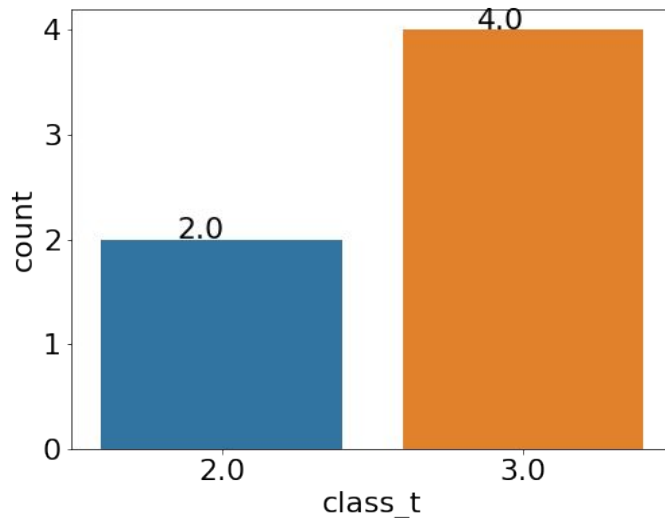
Preprocessing Overview & Results

-Cleaned the data and cross-matched against the truth catalogue :
Class 1: SS-AGN, Class 2: FS-AGN, Class 3: SFG

560MHz



9200MHz



ML Models performance - Random Forest

```
1 Image: ../data/sample_images/560mhz_1000h.fits, frequency: 560 MHz
2 Classifier: RandomForest
3 Score was 17330.711282542998
4 Number of detections 25996
5 Number of matches 25859
6 Number of matches too far from truth 95
7 Number of false detections 137
8 Score for all matches 17467.711282542998
9 Accuracy percentage 67.54983287266715
10 Classification report:
11           precision    recall  f1-score   support
12
13    1 (SS-AGN)      0.0000      0.0000      0.0000         303
14    2 (FS-AGN)      0.3402      0.2523      0.2897         329
15    3 (SFG)         0.9786      0.9937      0.9861        25227
16
17   accuracy                   0.9726        25859
18   macro avg              0.4396      0.4153      0.4253        25859
19   weighted avg           0.9590      0.9726      0.9657        25859
```


ML Models performance - Gradient Boosting

```
1 Image: ../data/sample_images/560mhz_1000h.fits, frequency: 560 MHz
2 Classifier: GradientBoosting
3 Score was 17338.282711114425
4 Number of detections 25996
5 Number of matches 25859
6 Number of matches too far from truth 95
7 Number of false detections 137
8 Score for all matches 17475.282711114425
9 Accuracy percentage 67.57911253766358
10 Classification report:
11           precision    recall  f1-score   support
12
13  1 (SS-AGN)      0.0000      0.0000      0.0000         303
14  2 (FS-AGN)      0.3500      0.0851      0.1369         329
15  3 (SFG)         0.9766      0.9979      0.9871       25227
16
17      accuracy                0.9746       25859
18      macro avg      0.4422      0.3610      0.3747       25859
19      weighted avg      0.9572      0.9746      0.9648       25859
```

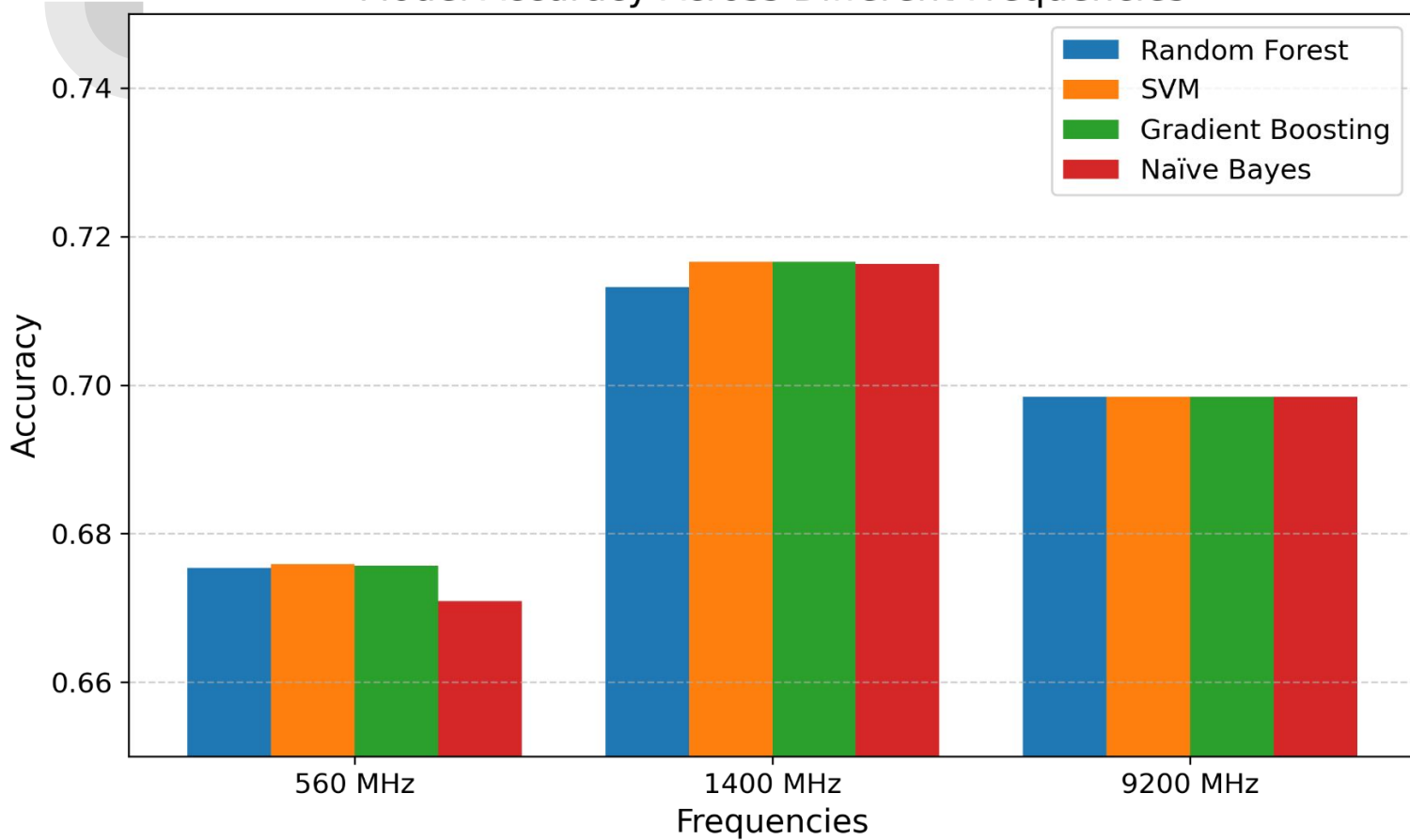
ML Models performance-NaiveBayes

```
1 |Image: ../data/sample_images/560mhz_1000h.fits, frequency: 560 MHz
2 Classifier: NaiveBayes
3 Score was 17212.854139685856
4 Number of detections 25996
5 Number of matches 25859
6 Number of matches too far from truth 95
7 Number of false detections 137
8 Score for all matches 17349.854139685856
9 Accuracy percentage 67.0940645024396
10 Classification report:
11           precision    recall  f1-score   support
12
13  1 (SS-AGN)      0.0000      0.0000      0.0000       303
14  2 (FS-AGN)      0.0043      0.0122      0.0063       329
15  3 (SFG)         0.9760      0.9641      0.9700    25227
16
17      accuracy          0.9407    25859
18      macro avg      0.3268      0.3254      0.3254    25859
19      weighted avg      0.9522      0.9407      0.9464    25859
```

ML Models performance - SVM

```
1 |Image: ../data/sample_images/560mhz_1000h.fits, frequency: 560 MHz
2 Classifier: SVM
3 Score was 17341.711282542998
4 Number of detections 25996
5 Number of matches 25859
6 Number of matches too far from truth 95
7 Number of false detections 137
8 Score for all matches 17478.711282542998
9 Accuracy percentage 67.59237125388839
10 Classification report:
11           precision    recall  f1-score   support
12
13  1 (SS-AGN)     0.0000     0.0000     0.0000       303
14  2 (FS-AGN)     0.0000     0.0000     0.0000       329
15  3 (SFG)        0.9756     1.0000     0.9876    25227
16
17      accuracy           0.9756    25859
18   macro avg       0.3252     0.3333     0.3292    25859
19 weighted avg       0.9517     0.9756     0.9635    25859
```

Model Accuracy Across Different Frequencies





Conclusion

We have:

- Performed primary beam correction and source finding for the three frequencies
- Explored and Pre_processed the data in preparation for training
- Trained different models to classify the sources
- Tested the performance of the models and found that SVM and Gradient boosting performed better than Random forest



Thank You

